Bertrand’s paradox

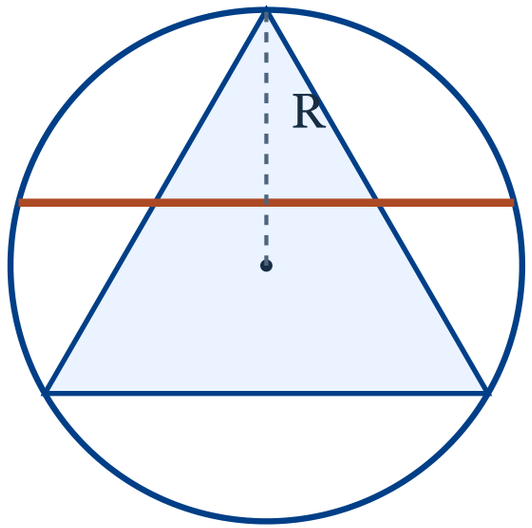
Supplementary · Bertrand’s paradox and sampling rules

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Random selection of a chord

A chord is selected at random in a circle. What is the probability that it is longer than the side of an inscribed equilateral triangle?



Triangle side = $\sqrt{3}$ R

Chord length = L

What is P(L > $\sqrt{3}$ R)?

We will calculate three answers to this same question, then make the probability models precise.

The chord-length question

For a circle of radius R, find the probability that a sampled chord has length $L > \sqrt{3}\ R$. What probability would you assign?

The reference length is $\sqrt{3}\ R$.

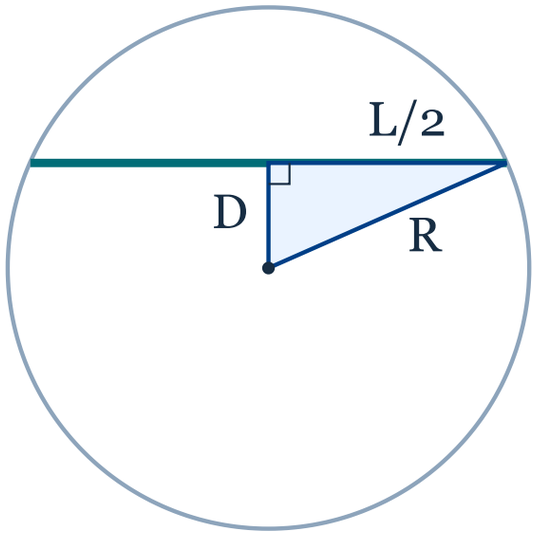
We will calculate the probability under three specified sampling rules.

In each case the required event is the same geometric condition:

$$L > \sqrt{3}\ R.$$

Chord length from the midpoint distance

Let D be the distance from the centre to the chord. Express its length L in terms of D.



$(L/2)^2 + D^2 = R^2$

Long $\Leftrightarrow$ $D < R/2$

The right triangle gives $(L/2)^2 + D^2 = R^2$.

Therefore $L = 2\sqrt{R^2 - D^2}$.

Converting the long-chord condition

Using $L = 2\sqrt{R^2 - D^2}$, express the strict condition $L > \sqrt{3}\ R$ as an inequality for D.

The perpendicular from the centre bisects the chord, giving

$$(L/2)^2 + D^2 = R^2, \qquad L = 2\sqrt{R^2 - D^2}\ .$$

Now apply the strict threshold:

$$2\sqrt{R^2 - D^2}\ > \sqrt{3}\ R.$$

Both sides are nonnegative, so squaring is valid:

$$4(R^2 - D^2) > 3R^2 \Leftrightarrow D^2 < R^2/4.$$

Since $D \geq 0$, the event is $D < R/2$.

A numerical geometry check

For $R = 10$ cm and $L = 2\sqrt{R^2 - D^2}$, calculate L when $D = 3$ cm and when $D = 6$ cm. Compare both lengths with $\sqrt{3}\ R$.

At $D = 3$ cm: $L = 2\sqrt{100 - 9} = 2\sqrt{91} \approx 19.08$ cm.

At $D = 6$ cm: $L = 2\sqrt{100 - 36} = 16$ cm.

The threshold is $10\sqrt{3} \approx 17.32$ cm.

The first chord is longer than the threshold; the second is shorter. This agrees with the test $D < 5$ cm.

Experiment I: a uniform angle at a fixed endpoint

Use a unit circle and fix $a = (1, 0)$. Choose θ uniformly in $(0, \pi)$, measured from the upward tangent towards the disk. Find the long-chord probability.

The resulting chord has length $L = 2\sin\theta$.

$$L > \sqrt{3} \iff \pi/3 < \theta < 2\pi/3.$$

The favorable interval occupies one third of the angular range:

$$\frac{2\pi/3 - \pi/3}{\pi} = \frac{1}{3}.$$

Experiment II: a uniform distance

Choose r uniformly in $(0,1)$. Set the midpoint to $(r,0)$ and draw the vertical chord through it. Find the long-chord probability.

$$L = 2\sqrt{1-r^2}\ , \qquad L > \sqrt{3} \ \Leftrightarrow r < 1/2.$$

Half the available distance interval is favorable:

$$\frac{1/2}{1} = \frac{1}{2}.$$

Fixing the midpoint direction specifies which chord is drawn.

Experiment III: a uniform midpoint

Choose a midpoint uniformly by area in the unit disk. Draw the chord perpendicular to the radius through that point. Find the long-chord probability.

The midpoint must lie inside the concentric disk of radius $1/2$.

$$\frac{\pi(1/2)^2}{\pi} = \frac{1}{4}.$$

The centre and boundary have area zero; omit them to make the construction unique and nondegenerate.

The same English question gives three answers

Meaning of “random chord”	Probability of $L > \sqrt{3}$
Uniform angle at a fixed endpoint	$1/3$
Uniform distance with a fixed direction	$1/2$
Uniform midpoint area	$1/4$

Each calculation follows its stated sampling rule.

We now make the outcomes, events and measures precise with $(\Omega, \mathcal{F}, P)$.

The angle experiment as a triplet

$$\Omega_1 = (0, \pi), \qquad \mathcal{F}_1 = \mathcal{B}((0, \pi)).$$

For a Borel event B, define

$$P_1(B) = \frac{1}{\pi} \int_B d\theta.$$

$$E_1 = (\pi/3, 2\pi/3), \qquad P_1(E_1) = \frac{1}{3}.$$

$\mathcal{B}$ denotes the Borel σ-algebra generated by the open sets.

The distance experiment as a triplet

$$\Omega_2 = (0,1), \qquad \mathcal{F}_2 = \mathcal{B}((0,1)).$$

For a Borel event B, define

$$P_2(B) = \int_B dr.$$

$$E_2 = (0,1/2), \qquad P_2(E_2) = \frac{1}{2}.$$

The fixed midpoint direction is part of the construction.

The midpoint experiment as a triplet

$$D^{*} = \{(x, y) : 0 < x^2 + y^2 < 1\}.$$

$$\Omega_3 = D^{*}, \qquad \mathcal{F}_3 = \mathcal{B}(D^{*}).$$

$$P_3(B) = \frac{1}{\pi} \iint_B dx\, dy.$$

$$E_3 = \{(x, y) : 0 < x^2 + y^2 < 1/4\}, \qquad P_3(E_3) = \frac{1}{4}.$$

The omitted centre and boundary have area zero.

The probability triplets resolve the contradiction

With the probability triplets specified, the apparent contradiction is resolved: the three answers belong to three different probability models.

$$P_1(E_1) = \frac{1}{3}, \qquad P_2(E_2) = \frac{1}{2}, \qquad P_3(E_3) = \frac{1}{4}.$$

Stopping point. The resolution is complete. The remaining slides study sampling maps and induced measures on a common chord space.

Exercise: connect the constructions

1. Randomize the first endpoint in Experiment I uniformly, independently of θ. Show that this gives two independent uniform endpoints.
2. Give Experiment II an independent uniform midpoint direction. Does its length distribution change?
3. Explain why an equal length distribution need not mean an equal measure on chords.

Exercise 8 in the supplement states the full problem.

In the advanced section, Model A uses independent uniform endpoints; B uses independent uniform distance and direction; C uses uniform midpoint area.

Coordinates that specify a chord

For radius R, let $t = D/R \in (0, 1)$ and $\phi \in [0, 2\pi)$ describe the midpoint. Let $C(\phi, t)$ be the chord through $Rt(\cos\phi, \sin\phi)$, perpendicular to that radius.

Let $t = D/R$ and let ϕ be the direction of the midpoint from the centre.

Place the midpoint at distance Rt in direction ϕ, then draw the perpendicular chord.

Call this construction $C(\phi, t)$.

We use $0 < t < 1$ and $0 \leq \phi < 2\pi$. Diameters and zero-length chords have probability zero under the three models.

Recovering the chord coordinates

For midpoint m and radius R, define $t = \| m \| /R$ and $\phi = \mathrm{Arg}(m) \in [0, 2\pi)$. Recover (ϕ, t) from the chord.

Measure its normalized midpoint distance and direction:

$$t = \frac{\| m \|}{R}, \qquad \phi = \mathrm{Arg}(m) \in [0, 2\pi).$$

Thus

$$C^{-1}(c) = (\phi, t).$$

This is an inverse function on the stated chord space. The supplement gives the endpoint formula and verifies both inverse identities.

Model B: uniform radial distance

Sample D uniformly on $(0, R)$ and independently ϕ uniformly on $[0, 2\pi)$. For $L = 2\sqrt{R^2 - D^2}$, find the probability of $L > \sqrt{3}\ R$.

The event is $D < R/2$.

The favourable radial interval has length $R/2$, out of total length R:

$$P_B(L > \sqrt{3}\ R) = \frac{R/2}{R} = \frac{1}{2}.$$

This model is uniform in radial distance.

Model B: sampling map and inverse

Let U, V be independent uniforms on $(0,1)$ and $[0,1)$, and set $t = U$, $\phi = 2\pi V$. Using $C(\phi, t)$, write the chord, recover (U, V), and express its length.

$$t = U, \qquad \phi = 2\pi V, \qquad c = C(2\pi V, U).$$

From the sampled chord, recover

$$U = t, \qquad V = \phi/(2\pi).$$

The length is $L_B = 2R\sqrt{1 - U^2}$, and the long-chord event is $U < 1/2$.

Model C: a midpoint uniform in disk area

Choose the chord midpoint uniformly by area in a disk of radius R. The long-chord event is D<R/2. Find the probability.

Long chords have midpoints within radius $R/2$.

Uniform area sampling gives

$$P_C(L > \sqrt{3}\ R) = \frac{\pi(R/2)^2}{\pi R^2} = \frac{1}{4}.$$

The smaller disk occupies a quarter of the total area.

Why uniform area uses a square root

For a midpoint sampled uniformly by area in a radius-R disk and U uniform on (0,1), derive P(D<r) and the corresponding transformation of U.

$$P(D < r) = \frac{\pi r^2}{\pi R^2} = \left(\frac{r}{R}\right)^2.$$

The transformation $D = R\sqrt{U}$ has exactly this property:

$$P(R\sqrt{U} < r) = P\left(U < (r/R)^2\right) = (r/R)^2.$$

Using $D = RU$ instead would give uniform radial distance.

Model C: sampling map and inverse

For uniform disk area, set $t = \sqrt{U}$, $\phi = 2\pi V$, with independent uniforms U, V on $(0, 1)$ and $[0, 1)$. Using $C(\phi, t)$, write the chord, its inverse and its length.

$$t = \sqrt{U}\ , \qquad \phi = 2\pi V, \qquad c = C(2\pi V, \sqrt{U}\ ).$$

Inverse: $U = t^2$, $V = \phi/(2\pi)$.

The length is $L_C = 2R\sqrt{1 - U}$.

The long-chord event is $U < 1/4$, which has probability $1/4$.

Practice: a smaller midpoint region

For the same disk of radius R, find P(D<R/3) under (i) a uniform radial distance and (ii) a midpoint chosen uniformly by disk area.

Model B, uniform radial distance:

$$P_B(D < R/3) = \frac{R/3}{R} = \frac{1}{3}.$$

Model C, uniform area:

$$P_C(D < R/3) = \frac{\pi(R/3)^2}{\pi R^2} = \frac{1}{9}.$$

The region is identical. The two sampling distributions assign it different probabilities.

Model A: two uniform endpoints

Choose independent uniform angles A_1, A_2 on $[0, 2\pi)$, and set $\Theta = (A_2 - A_1) \bmod 2\pi$. Express the chord length using Θ.

After the first angle is chosen, rotational symmetry leaves Θ uniform on $[0, 2\pi)$.

Bisect the isosceles triangle formed by the centre and the two endpoints. Then

$$L/2 = R\sin(\Theta/2), \qquad L = 2R\sin(\Theta/2).$$

For $\Theta = \pi$, the endpoints are opposite and $L = 2R$.

Model A: the favourable angular interval

Let $\Theta = (A_2 - A_1) \bmod 2\pi$ be uniform on $[0, 2\pi)$ and $L = 2R\sin(\Theta/2)$. Solve $L > \sqrt{3}\ R$ for Θ.

$$\sin(\Theta/2) > \sqrt{3}\ /2.$$

Since $0 \le \Theta/2 < \pi$,

$$\pi/3 < \Theta/2 < 2\pi/3.$$

Therefore $2\pi/3 < \Theta < 4\pi/3$, and

$$P_A(\text{long}) = \frac{4\pi/3 - 2\pi/3}{2\pi} = \frac{1}{3}.$$

Model A in midpoint coordinates

For uniform Θ on $[0, 2\pi)$, define $\alpha = \min(\Theta, 2\pi - \Theta)$. Find its distribution and the midpoint distance $t = D/R$.

$$\alpha = \min(\Theta, 2\pi - \Theta), \qquad 0 \le \alpha \le \pi.$$

The right triangle gives $t = D/R = \cos(\alpha/2)$.

For $0 \le a \le \pi$, the angular intervals producing $\alpha \le a$ have total length $2a$:

$$P(\alpha \le a) = \frac{2a}{2\pi} = \frac{a}{\pi}.$$

Thus α is uniform on $[0, \pi]$.

Model A: sampling map and inverse

For independent uniform endpoint angles, set $U = \alpha/\pi$, $\alpha = \min(\Theta, 2\pi - \Theta)$ and $V = \phi/(2\pi)$. With $t = \cos(\alpha/2)$, find the chord map and its inverse.

$$t = \cos(\pi U/2), \qquad c = C(2\pi V, \cos(\pi U/2)).$$

Inverse:

$$U = \frac{2}{\pi}\arccos(t), \qquad V = \phi/(2\pi).$$

This generates the same distribution of unordered chords as independent uniform endpoints. The supplement proves the independence of the midpoint direction and the smaller separation.

Model A: the length random variable

For independent uniform endpoint angles, use $U = \alpha/\pi$, $t = \cos(\alpha/2)$. Express $L = 2R\sqrt{1-t^2}$ in terms of U and find the condition for $L > \sqrt{3}\ R$.

$$L_A = 2R\sqrt{1-\cos^2(\pi U/2)} = 2R\sin(\pi U/2).$$

The sine is increasing for $0 < U < 1$, so

$$L_A > \sqrt{3}\ R \Leftrightarrow U > 2/3.$$

Therefore $P_A(\text{long}) = 1 - 2/3 = 1/3$.

Practice: recovering the source coordinates

For $t = 1/2$ and $\phi = \pi$, recover $v = \phi/(2\pi)$ and u under Model A, $t = \cos(\pi U/2)$; Model B, $t = U$; and Model C, $t = \sqrt{U}$. Is the chord longer than $\sqrt{3}\ R$?

In every model, $v = \phi/(2\pi) = 1/2$.

Model A: $u = (2/\pi)\arccos(1/2) = 2/3$.

Model B: $u = t = 1/2$.

Model C: $u = t^2 = 1/4$.

The chord length equals $\sqrt{3}\ R$, so it is outside the strict long-chord event in all three models.

Comparing the three sampling distributions

Summarize the event calculations using the same uniform source U.

Sampling rule	Normalized distance t	Long-chord condition	Probability
A: uniform endpoints	$\cos(\pi U/2)$	$U > 2/3$	$1/3$
B: uniform radial distance	U	$U < 1/2$	$1/2$
C: uniform midpoint area	$\sqrt{U}$	$U < 1/4$	$1/4$

Each result uses a specified random variable and a specified sampling distribution.

Three measures on a common chord space

How can the rotationally invariant models use the same chord space?

Let $\mathcal{C}_*$ be the nondegenerate, nondiameter chords and let $\mathcal{F}_{\mathcal{C}_*}$ be the event collection defined in the supplement.

$$(\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_A), \quad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_B), \quad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_C).$$

The event is $E = \{c : D(c) < R/2\}$ in all three models.

The sampling rules assign different probabilities to regions of the chord space.

“Random chord” does not select one of these probability distributions.

Compare the sampling experiments · Inspect the maps and inverses

Changing coordinates in one fixed model

Keep Model B, where $T = D/R$ is uniform on $(0, 1)$. Put $W = T^2$. What is $P(W < 1/4)$?

$$W < 1/4 \Leftrightarrow T^2 < 1/4 \Leftrightarrow T < 1/2.$$

Hence $P_B(W < 1/4) = 1/2$.

W is not uniform: in general, $P_B(W < w) = \sqrt{w}$ for $0 < w < 1$.

Declaring W uniform would change the sampling model to Model C.